Advanced Material Physics: Exercises

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Unit system used in these exercises. Throughout these exercises, $4\pi\epsilon_0 = 1$.

Scope of the exercises. The exercises follow the topics of Part 1 of R. Coelho, *Physics of Dielectrics for the Engineer* (Elsevier, 1979), up to the Onsager model (Sec. V.3). They are written in the notation and the units of the lecture note *Advanced Material Physics*, with $4\pi\epsilon_0 = 1$, and $1\ \mathrm{D} = 10^{-18}$ statC cm. Each exercise names the related sections of the note and of the book, and ends with an answer for checking.

1 Multipole Expansion

Exercise 1: Origin dependence of the moments. A charge $3q$ sits at $z = +d$ and a charge q at $z = -d$.

(a) Find the monopole moment Q and the dipole moment $\vec{p}$ about the midpoint.

(b) Find the origin on the z axis about which the dipole moment vanishes.

(c) Find the quadrupole component $\mathcal{Q}_{zz}$ about the midpoint and about the origin of part (b).

Answer. (a) $Q = 4q$ and $\vec{p} = 2qd\,\hat{z}$. (b) $z = d/2$, the charge centroid $\sum_i q_i z_i / Q$. (c) $\mathcal{Q}_{zz} = 8qd^2$ about the midpoint and $6qd^2$ about $z = d/2$.
Related: Section 1 of the note; Coelho, Sec. II.1.

Exercise 2: Planar quadrupole. Charges $+q$ sit at $(\pm a, 0, 0)$ and charges $-q$ at $(0, \pm a, 0)$.

(a) Show that $Q = 0$ and $\vec{p} = 0$.

(b) Compute the quadrupole moment tensor $\mathcal{Q}$ of Section 1 of the note and check that its trace vanishes.

(c) Write the leading term of the far-field potential in terms of the polar angle θ and the azimuthal angle φ of $\hat{R}$.

Answer. (b) $\mathcal{Q} = \mathrm{diag}(6qa^2, -6qa^2, 0)$. (c) $\phi \approx 3qa^2 \sin^2\theta \cos 2\varphi / R^3$.
Related: Section 1 of the note; Coelho, Sec. II.1.

Exercise 3: Legendre expansion of a physical dipole.

(a) Show that the first three terms of

$$\frac{1}{|\vec{R} - \vec{r}|} = \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} \mathcal{P}_l(\cos\gamma),$$

where γ is the angle between $\vec{R}$ and $\vec{r}$, reproduce the expansion of Section 1 of the note.

(b) Charges $\pm q$ sit at $\pm(s/2)\hat{z}$. Using $\mathcal{P}_l(-x) = (-1)^l \mathcal{P}_l(x)$, show that only odd l contribute to the potential.

(c) Find the ratio of the $l = 3$ term to the $l = 1$ term.

Answer. (b) $\phi = 2q \sum_{l\ \mathrm{odd}} (s/2)^l \mathcal{P}_l(\cos\theta)/R^{l+1}$, with the point dipole $p = qs$ as the leading term and no quadrupole. (c) $(s/2R)^2 \mathcal{P}_3(\cos\theta)/\mathcal{P}_1(\cos\theta)$.
Related: Sections 1 and 3 of the note; Coelho, Secs. II.2–II.3.

Exercise 4: Partial charges from measured dipole moments. The gas-phase HCl molecule has the bond length $L = 1.275$ Å and the dipole moment 1.08 D. The water molecule has the O–H bond length $b = 0.958$ Å, the H–O–H angle $\psi = 104.5°$, and the dipole moment 1.85 D.

(a) Model HCl as the charges $\pm\delta e$ on the two nuclei. Find δ, which is the fraction of ionic character p/eL.

(b) Model water as the charge $+\delta e$ on each H and $-2\delta e$ on O. Show that the dipole moment is $p = 2\delta e\, b \cos(\psi/2)$, directed along the bisector of the H–O–H angle, and find δ.

(c) Find the dipole moment of a fully ionic water molecule, $\delta = 1$, and compare it with the measured value.

Answer. (a) $eL = 6.12$ D and $\delta = 0.18$. (b) $\delta = 0.33$. (c) 5.63 D, three times the measured value, so the O–H bonds are mostly covalent.
Related: Section 1 of the note; Coelho, Sec. III.3.

2 Induced Polarizability

Models for the atomic polarizability. Section 4 of the note defines the atomic polarizability $\alpha_{\rm pol}$ but does not model it. Exercises 5–13 build simple models and connect them to the local field.

Exercise 5: Atom as a nucleus in a uniform electron cloud. A nucleus of charge Ze sits at the center of a rigid, uniformly charged electron cloud of radius R and total charge $-Ze$. A field $\vec{E}$ displaces the nucleus by $d \ll R$.

(a) Use Gauss's law for the field of the cloud to find the restoring force on the nucleus. Show that the induced dipole moment is $\vec{p} = R^3\vec{E}$, so that $\alpha_{\rm pol} = R^3$.

(b) For $Z = 1$, $R = 1$ Å, and $E = 3 \times 10^7$ V/m, find d and d/R.

Answer. (a) The cloud field at distance d from its center is Zed/R^3, so $Ze\,E = Z^2e^2d/R^3$. (b) $E = 1.0 \times 10^3$ statV/cm, $d = 2.1 \times 10^{-12}$ cm, and $d/R = 2.1 \times 10^{-4}$, so the linear response holds.
Related: Section 4 of the note; Coelho, Sec. III.2.

Exercise 6: Atom as a dielectric sphere.

(a) Model an atom as a sphere of radius R and dielectric constant ε_s in vacuum. From the effective dipole moment of the dielectric sphere in Section 3 of the note, with $\varepsilon_1 = \varepsilon_s$ and $\varepsilon_2 = 1$, find $\alpha_{\rm pol}$. Check the limits $\varepsilon_s \to 1$ and $\varepsilon_s \to \infty$.

(b) A material consists of such spheres with the number density n, which fill the volume fraction $f_V = \frac{4\pi}{3}R^3 n$. Combine part (a) with the Clausius–Mossotti relation to find ε of the material.

Answer. (a) $\alpha_{\rm pol} = R^3(\varepsilon_s - 1)/(\varepsilon_s + 2)$. The limit $\varepsilon_s \to \infty$ gives R^3, as in Exercise 5. (b) $(\varepsilon - 1)/(\varepsilon + 2) = f_V(\varepsilon_s - 1)/(\varepsilon_s + 2)$, the Maxwell Garnett mixing formula in a vacuum host.
Related: Sections 3 and 4 of the note; Coelho, Sec. III.2.

Exercise 7: Permittivity of packed polarizable spheres. Conducting spheres of radius R, each with $\alpha_{\rm pol} = R^3$, fill space in a simple cubic packing ($f_V = \pi/6$) or a face-centered cubic packing ($f_V = \pi\sqrt{2}/6$), where the spheres touch.

(a) Neglect the local-field correction, $\vec{E}_{\rm loc} = \vec{E}$, and show that $\varepsilon = 1 + 4\pi n\alpha_{\rm pol} = 1 + 3f_V$. Evaluate ε for both packings.

(b) Use the Clausius–Mossotti relation instead and evaluate ε for both packings.

(c) Explain why the two results differ more for the denser packing.

Answer. (a) 2.57 for simple cubic and 3.22 for face-centered cubic packing. (b) $\varepsilon = (1+2f_V)/(1-f_V)$ gives 4.30 and 9.56. (c) The local-field correction grows with the polarization, and Clausius–Mossotti diverges as $f_V \to 1$.
Related: Section 4 of the note; Coelho, Sec. III.2.

Exercise 8: Polarizability of a diatomic molecule. Two identical atoms of polarizability $\alpha_{\rm pol}$ sit at the distance L. Each atom feels the external field plus the point-dipole field of the other atom (Section 2 of the note).

(a) For a field along the molecular axis, show that the molecular polarizability is $\alpha_\parallel = 2\alpha_{\rm pol}/(1 - 2\alpha_{\rm pol}/L^3)$.

(b) For a field perpendicular to the axis, show that $\alpha_\perp = 2\alpha_{\rm pol}/(1 + \alpha_{\rm pol}/L^3)$.

(c) With $\alpha_{\rm pol} = R^3$ (Exercise 5) and touching atoms, $L = 2R$, evaluate $\alpha_\parallel$, $\alpha_\perp$, and the anisotropy $\Delta\alpha = \alpha_\parallel - \alpha_\perp$.

Answer. (c) $\alpha_\parallel = \frac{8}{3}\alpha_{\rm pol}$, $\alpha_\perp = \frac{16}{9}\alpha_{\rm pol}$, and $\Delta\alpha = \frac{8}{9}\alpha_{\rm pol} > 0$. An elongated molecule is easier to polarize along its axis.
Related: Sections 2 and 4 of the note; Coelho, Sec. III.3.

Exercise 9: Polarizable-ion model of the NaCl molecule. A gas-phase NaCl molecule consists of the ions Na^+ and Cl^- at the distance $L = 2.361$ Å. The ions have the polarizabilities $\alpha_+ = 0.41$ Å^3 and $\alpha_- = 2.96$ Å^3. The measured dipole moment of the molecule is 9.0 D.

(a) Show that the charge of each ion polarizes the other ion, and that both induced dipoles point opposite to the ionic dipole of magnitude eL.

(b) Neglect the fields of the induced dipoles and show that the dipole moment of the molecule is $p = eL - e(\alpha_+ + \alpha_-)/L^2$. Evaluate p.

(c) Include the point-dipole field of each induced dipole at the other ion (Section 2 of the note). Show that the induced dipoles satisfy

$$p_+ = \alpha_+ \left(\frac{e}{L^2} + \frac{2p_-}{L^3} \right), \qquad p_- = \alpha_- \left(\frac{e}{L^2} + \frac{2p_+}{L^3} \right).$$

Solve for $p_+ + p_-$ and compare the resulting dipole moment with the measured value.

Answer. (b) $eL = 11.34$ D, the induced dipoles reduce it by 2.90 D, and $p = 8.44$ D. (c) $p_+ + p_- = (e/L^2)\left(\alpha_+ + \alpha_- + 4\alpha_+\alpha_-/L^3\right) / \left(1 - 4\alpha_+\alpha_-/L^6\right) = 3.32$ D, so that $p = 8.03$ D. The model underestimates the measured 9.0 D, because the point-dipole approximation is crude for ions in contact.
Related: Section 2 of the note; Coelho, Sec. III.3.

Exercise 10: Symmetry of the polarizability tensor. An anisotropic molecule has the induced dipole moment $p_\beta = \sum_\gamma \alpha^{\rm pol}_{\beta\gamma} E_\gamma$. When the field changes by $d\vec{E}$, the energy of the induced dipole changes by $dw = -\sum_\beta p_\beta \, dE_\beta$.

(a) Require dw to be an exact differential and show that $\alpha^{\rm pol}_{\beta\gamma} = \alpha^{\rm pol}_{\gamma\beta}$.

(b) Explain why a rotation to the principal axes makes the tensor diagonal, with three principal polarizabilities.

(c) Apply the same argument to the energy density of a dielectric to show that the dielectric tensor $\varepsilon_{\beta\gamma}$ of Section 3 of the note is symmetric.

Answer. (a) The condition $\partial(\sum_\gamma \alpha^{\rm pol}_{\beta\gamma} E_\gamma)/\partial E_\delta = \partial(\sum_\gamma \alpha^{\rm pol}_{\delta\gamma} E_\gamma)/\partial E_\beta$ gives $\alpha^{\rm pol}_{\beta\delta} = \alpha^{\rm pol}_{\delta\beta}$. (b) A real symmetric matrix has real eigenvalues and orthogonal eigenvectors.
Related: Section 3 of the note; Coelho, Sec. III.3.

Exercise 11: Orientation average of an anisotropic polarizability. A molecule has the principal polarizabilities α_1, α_2, and α_3. Its orientation is random and is not affected by the field.

(a) Show that the average component of the induced dipole along the field is $\langle \vec{p} \cdot \hat{E} \rangle = \bar{\alpha} E$, with $\bar{\alpha} = (\alpha_1 + \alpha_2 + \alpha_3)/3$. Use $\langle u_i^2 \rangle = 1/3$ for the components of a random unit vector.

(b) A molecule is polarizable along one axis only, with $\alpha_1 = R^3$ and $\alpha_2 = \alpha_3 = 0$. Find $\bar{\alpha}$ and compare it with a molecule whose axis is held parallel to the field.

Answer. (a) $\vec{p} \cdot \hat{E} = \sum_i \alpha_i (\hat{u}_i \cdot \hat{E})^2 E$, and each square averages to 1/3. (b) $\bar{\alpha} = R^3/3$, a third of the aligned value R^3.
Related: Section 3 of the note; Coelho, Secs. III.2–III.3.

Exercise 12: Ionic polarizability. Two ions of charges $\pm e$ and reduced mass m_r are bound by a spring of constant κ, so that their vibration frequency is $\omega_0 = \sqrt{\kappa/m_r}$.

(a) A static field $\vec{E}$ along the bond stretches it by x. Show that the ionic polarizability is $\alpha_{\text{ion}} = e^2/(m_r \omega_0^2)$.

(b) For NaCl, use the atomic masses 23.0 u and 35.5 u and the transverse optical frequency $\omega_0/2\pi c = 164$ cm^{-1} to estimate α_{ion}. Compare it with $\alpha_{\text{pol}} = R^3$ for $R = 1$ Å.

Answer. (b) $m_r = 2.32 \times 10^{-23}$ g, $\omega_0 = 3.09 \times 10^{13}$ s^{-1}, and $\alpha_{\text{ion}} = 1.0 \times 10^{-23}$ cm^3 $= 10$ Å^3, an order of magnitude larger than the electronic estimate.
Related: Section 4 of the note; Coelho, Sec. III.4.

Exercise 13: Near-field sum of a dipole chain. An infinite chain of equal dipoles $\vec{p}$ has the spacing a. Use the point-dipole field of Section 2 of the note and $\zeta(3) = \sum_{n=1}^{\infty} n^{-3} \approx 1.202$.

(a) Find the field at one site produced by all other dipoles, for dipoles along the chain.

(b) Repeat part (a) for dipoles perpendicular to the chain.

(c) The Lorentz model of Section 4 of the note sets $\vec{E}_{\text{near}} = 0$. Explain why this assumption holds for a cubic lattice but fails for a chain.

(d) The chain now consists of atoms of polarizability α_{pol}, polarized by an external field. Use parts (a) and (b) to find the polarizability per atom for a field along and perpendicular to the chain. Evaluate both for $\alpha_{\text{pol}} = R^3$ and touching atoms, $a = 2R$.

Answer. (a) $4\zeta(3)\,\vec{p}/a^3 \approx 4.81\,\vec{p}/a^3$. (b) $-2\zeta(3)\,\vec{p}/a^3 \approx -2.40\,\vec{p}/a^3$. (c) A chain lacks the equivalence of the three axes that makes $\sum_k (3z_k^2 - r_k^2)/r_k^5$ vanish. (d) $\alpha_{\mathrm{pol}}/(1 - 4\zeta(3)\alpha_{\mathrm{pol}}/a^3) = 2.51\,\alpha_{\mathrm{pol}}$ along the chain and $\alpha_{\mathrm{pol}}/(1 + 2\zeta(3)\alpha_{\mathrm{pol}}/a^3) = 0.77\,\alpha_{\mathrm{pol}}$ perpendicular to it.
Related: Section 4 of the note; Coelho, Sec. III.4 (problem on the one-dimensional lattice).

3 Orientational Polarization

Exercise 14: Langevin function beyond the linear term.

(a) Expand $e^{\xi \cos\vartheta}$ in powers of ξ before integrating the partition function of Section 4.3 of the note. Find $\ln Z$ to order ξ^4 and show that $\mathcal{L}(\xi) = \xi/3 - \xi^3/45 + \mathcal{O}(\xi^5)$.

(b) For $p_0 = 1.85$ D at $T = 300$ K, find the field at which $\langle p_{0,z} \rangle$ falls 1% below the linear law.

(c) Evaluate the orientational polarizability α_{or} for $p_0 = 1$ D at 300 K and compare it with the electronic estimate R^3 for $R = 1$ Å.

Answer. (a) $\ln Z = \ln 4\pi + \xi^2/6 - \xi^4/180 + \mathcal{O}(\xi^6)$. (b) $\xi^2/15 = 0.01$ gives $\xi = 0.387$ and $E = 8.7 \times 10^3$ statV/cm $= 2.6 \times 10^8$ V/m, far above ordinary laboratory fields. (c) $\alpha_{\mathrm{or}} = 8.0$ Å^3, about eight times the electronic estimate.
Related: Section 4.3 of the note; Coelho, Sec. IV.1.

Exercise 15: Dipoles with discrete orientations and spontaneous polarization.

(a) Dipoles of magnitude p_0 can point only along $\pm\hat{z}$, parallel to the field. Show that $\langle p_{0,z} \rangle = p_0 \tanh\xi$, with $\xi = p_0 E/k_B T$.

(b) In a cubic crystal, dipoles can point along the eight body diagonals $\langle 111 \rangle$. For a field along [001], find $\langle p_{0,z} \rangle$ and its low-field limit. Compare with the Langevin result.

(c) A set of N allowed orientations $\hat{u}_k$ has cubic symmetry. Show that $\sum_k \hat{u}_k \hat{u}_k^T = (N/3)\, I$, where I is the unit matrix. Conclude that the low-field response $\langle \vec{p}_0 \cdot \hat{E} \rangle = p_0 \xi/3$ holds for any direction of the field.

(d) Let the dipoles of part (a) feel the local field $E + \gamma P$, with $P = n\langle p_{0,z} \rangle$. Show that a spontaneous polarization appears below $k_B T_c = \gamma n p_0^2$. Repeat for freely rotating dipoles and compare with the critical temperature of Section 4 of the note for $\gamma = 4\pi/3$ and $\alpha_{\mathrm{pol}} = 0$.

Answer. (b) $\langle p_{0,z}\rangle = (p_0/\sqrt{3})\tanh(\xi/\sqrt{3}) \approx p_0\xi/3$, the same as for free dipoles, because the cubic symmetry makes the low-field response isotropic. (c) The matrix commutes with all cubic rotations, so it is proportional to I, and its trace is N. To first order, $\langle \vec{p}_0 \cdot \hat{E}\rangle = (p_0\xi/N)\sum_k (\hat{u}_k \cdot \hat{E})^2 = p_0\xi/3$. (d) Free dipoles give $k_B T_c = \gamma n p_0^2/3 = 4\pi n p_0^2/9$, the Debye critical temperature with $\alpha_{\rm pol} = 0$.
Related: Sections 4.3 and 4.4 of the note; Coelho, Sec. IV.2.

Exercise 16: Impurity–vacancy dipole in a crystal. In a simple cubic lattice of constant a, a divalent impurity binds a vacancy on one of its six nearest-neighbor sites. The pair forms a dipole of magnitude $p_0 = ea$. A field $\vec{E}$ points along [001].

(a) Find the energies of the six vacancy positions and show that

$$\langle p_{0,z}\rangle = p_0\,\frac{e^{\xi} - e^{-\xi}}{e^{\xi} + e^{-\xi} + 4}, \qquad \xi = \frac{p_0 E}{k_B T}.$$

(b) Show that the low-field limit is the Langevin result $p_0^2 E/3k_B T$.

(c) For a dilute concentration n of such pairs, find their contribution to ε.

Answer. (a) Two positions have the energies $\mp p_0 E$, and four have zero energy. (b) $\langle p_{0,z}\rangle = p_0\xi/3 + \mathcal{O}(\xi^5)$. (c) $\Delta\varepsilon = 4\pi n p_0^2/3k_B T$.
Related: Section 4.3 of the note; Coelho, Sec. IV.2.

Exercise 17: Anisotropic polar molecule. A linear molecule carries the permanent dipole $\vec{p}_0$ along its axis and has the polarizabilities $\alpha_\parallel$ along the axis and $\alpha_\perp$ perpendicular to it, with $\Delta\alpha = \alpha_\parallel - \alpha_\perp > 0$. The axis makes the angle ϑ with the field.

(a) Show that the orientation energy is $w(\vartheta) = -p_0 E\cos\vartheta - \frac{1}{2}\left(\Delta\alpha\cos^2\vartheta + \alpha_\perp\right)E^2$.

(b) Find the equilibrium orientations. Show that the antiparallel orientation $\vartheta = \pi$ becomes stable for $E > p_0/\Delta\alpha$.

(c) For a polar molecule, expand the Boltzmann average of $p_E = p_0\cos\vartheta + (\Delta\alpha\cos^2\vartheta + \alpha_\perp)E$ to third order in E. With $\beta = \Delta\alpha\, k_B T/p_0^2$, show that the cubic term is $-(1 - 4\beta - 2\beta^2)\, p_0^4 E^3/45 k_B^3 T^3$. Find the value β_0 above which the polarization grows faster than linearly (antisaturation).

(d) For a nonpolar molecule, $p_0 = 0$, show that at low fields $\langle\cos^2\vartheta\rangle = \frac{1}{3} + \frac{4}{45}z$, with $z = \Delta\alpha E^2/2k_B T$. Find the order parameter $S = \frac{1}{2}(3\langle\cos^2\vartheta\rangle - 1)$.

Answer. (b) $dw/d\vartheta = E\sin\vartheta\,(p_0 + \Delta\alpha E\cos\vartheta)$, and $d^2w/d\vartheta^2 = E(\Delta\alpha E - p_0)$ at $\vartheta = \pi$. (c) The linear term is $\left(\bar{\alpha} + p_0^2/3k_BT\right)E$, with $\bar{\alpha} = (\alpha_\parallel + 2\alpha_\perp)/3$, and $\beta_0 = \sqrt{6}/2 - 1 \approx 0.225$. (d) $S = \Delta\alpha E^2/15k_BT$.
Related: Section 4.3 of the note; Coelho, Secs. III.3 and IV.3.

Exercise 18: Electrocaloric effect. An isotropic dielectric with the temperature-dependent dielectric constant $\varepsilon(T)$ is charged to the field E at constant temperature. Per unit volume, the potential $G(T, E)$ satisfies $dG = -s\,dT - (D/4\pi)\,dE$, where s is the entropy density.

(a) Derive the Maxwell relation $(\partial s/\partial E)_T = (1/4\pi)(\partial D/\partial T)_E$.

(b) Show that the entropy change is $\Delta s = (E^2/8\pi)\,\partial\varepsilon/\partial T$.

(c) For a dilute polar material with $\varepsilon = \varepsilon_\infty + 4\pi n p_0^2/3k_BT$, find the heat $Q = T\Delta s$ absorbed per unit volume and interpret its sign.

Answer. (c) $Q = -np_0^2E^2/6k_BT$. The material releases heat, because the field orders the dipoles and lowers the entropy.
Related: Section 4.3 of the note; Coelho, Sec. IV.1.

4 Local-Field Models

Exercise 19: Dipole moment from a Debye plot. A polar gas at the fixed number density $n = 2.69 \times 10^{19}$ cm^{-3} has $\varepsilon - 1 = 4.17 \times 10^{-3}$ at 300 K and 3.35×10^{-3} at 400 K.

(a) Show that the Debye equation reduces to $\varepsilon - 1 = 4\pi n\left(\alpha_{\mathrm{pol}} + p_0^2/3k_BT\right)$ for $\varepsilon - 1 \ll 1$.

(b) Find p_0 and α_{pol}.

Answer. (b) $p_0 = 1.10$ D and $\alpha_{\mathrm{pol}} = 2.6$ Å^3.
Related: Section 4.3 of the note; Coelho, Sec. V.1.

Exercise 20: Local-field correction factors and the Kirkwood factor. Write the orientational contribution as $\varepsilon_s - \varepsilon_\infty = A \cdot 4\pi n p_0^2/3k_BT$, where ε_s is the static dielectric constant, ε_∞ is the optical dielectric constant, and A is a local-field correction factor.

(a) From the Debye equation, with ε_∞ given by the Clausius–Mossotti relation, show that $A_{\mathrm{CM}} = (\varepsilon_s + 2)(\varepsilon_\infty + 2)/9$.

(b) From the Onsager equation, show that $A_{\mathrm{O}} = \varepsilon_s(\varepsilon_\infty + 2)^2/[3(2\varepsilon_s + \varepsilon_\infty)]$.

(c) Show that both factors tend to one for a dilute material.

(d) For water ($\varepsilon_s = 78.4$, $\varepsilon_\infty = 1.78$, $n = 3.34 \times 10^{22}$ cm^{-3}, $T = 298$ K), compute A_{CM} and A_{O} and the dipole moment that each model infers from ε_s. Compare with the gas-phase value $p_0 = 1.85$ D.

(e) Replace p_0^2 by $g\,p_0^2$ in the Onsager equation and find the Kirkwood correlation factor g of water.

Answer. (d) $A_{\mathrm{CM}} = 33.8$ and $A_{\mathrm{O}} = 2.36$. The inferred dipole moments are 4.75 D for $A = 1$, 0.82 D for A_{CM}, and 3.09 D for A_{O}. (e) $g = 2.80$, which reflects the alignment of neighboring molecules by hydrogen bonds.
Related: Section 4 of the note; Coelho, Secs. V.2–V.3.

Exercise 21: Limits and temperature dependence of the Onsager model.

(a) Start from the self-consistent dipole moment and the directing field of the Onsager model (Section 4.4 of the note). Set $p_0 = 0$, use $\frac{4\pi}{3}R_c^3 n = 1$, and show that the Onsager model reproduces the Clausius–Mossotti relation.

(b) Solve the Onsager equation for ε as the positive root of a quadratic equation.

(c) For water ($\varepsilon_\infty = 1.78$, $4\pi n p_0^2/9k_B = 1156$ K, with n held fixed), evaluate ε at 273 K, 298 K, and 373 K. Compare with the measured values 87.9, 78.4, and 55.6.

Answer. (b) $\varepsilon = \frac{1}{4}\left[\varepsilon_\infty + Kc + \sqrt{(\varepsilon_\infty + Kc)^2 + 8\varepsilon_\infty^2}\right]$, with $K = 4\pi n p_0^2/9k_BT$ and $c = (\varepsilon_\infty + 2)^2$. (c) 31.2, 28.7, and 23.1. The model underestimates the measured values by a factor that decreases from 2.8 to 2.4 as the temperature rises.
Related: Section 4.4 of the note; Coelho, Sec. V.3.